

Kenneth Oswin

Software Engineer

Artificial Intelligence • Software Systems • Backend Development

Bangalore, Karnataka, India

+91 9632584998

kennethoswin289@gmail.com

Portfolio • GitHub • LinkedIn • LeetCode

Education

RV Institute of Technology and Management

2021 – 2025

Bachelor of Engineering in Information Science and Engineering (VTU)

CGPA: **8.36** / 10

St. Joseph's Pre-University College

2019 – 2021

Pre-University Education (State Board)

Percentage: **94%**

Bishop Cotton Boys' School

2007 – 2019

Indian Certificate of Secondary Education (ICSE)

Percentage: **94.6%**

Profile

Software Engineer with professional experience developing Python libraries, reusable software frameworks, backend services, and engineering tools for enterprise products. Beyond industry work, actively designing AI-powered software systems focused on autonomous agents, natural language driven software engineering, and intelligent developer tooling. Interested in pursuing graduate research in Artificial Intelligence, Software Engineering, and Large Language Model based systems.

Professional Experience

Crestron

July 2025 – Present

Associate Software Development Engineer in Test

- Develop reusable Python libraries and Robot Framework components supporting automation and software validation across multiple enterprise products.
- Designed a complete Robot Framework library from scratch for an unreleased hardware platform, abstracting low-level hardware interactions into reusable software components.
- Develop backend REST APIs enabling hardware control beyond framework limitations, significantly reducing manual engineering effort.
- Engineered CI/CD workflows using Jenkins and Docker to continuously execute engineering frameworks across multiple environments.
- Developed internal dashboards, execution trackers, and supporting software utilities for monitoring framework execution and engineering productivity.
- Worked across three independent enterprise products, adapting framework architecture, software libraries, and backend services to evolving technical requirements.
- Collaborated closely with software developers throughout feature development and system integration, improving maintainability and engineering efficiency.
- Mentored two engineering interns in Python development, framework architecture, and software engineering practices, both of whom transitioned into full-time roles.

- Developed Python-based software libraries and Robot Framework components supporting enterprise engineering workflows.
- Integrated engineering frameworks with Jenkins and Jira to improve continuous execution and engineering collaboration.
- Collaborated with cross-functional software teams to improve framework quality, maintainability, and engineering productivity.

Research & Engineering Projects

KenBot OS — Autonomous AI Operating System

2025 – Present

- Designed and developed an autonomous AI operating system that integrates Large Language Models, browser automation, scheduling services, persistent memory, and backend APIs into a unified intelligent software platform.
- Architected a modular service-oriented backend using Python and Flask, enabling independent AI services for conversational reasoning, content generation, task scheduling, analytics, and platform orchestration.
- Implemented persistent conversational memory incorporating contextual history, adaptive user profiling, relationship modelling, and long-term knowledge retention to improve personalized interactions across multiple sessions.
- Developed an AI orchestration layer coordinating Claude AI, browser automation, external APIs, scheduled jobs, and background workers to automate complex multi-step workflows.
- Designed reusable software modules for prompt management, service communication, monitoring, logging, analytics, and workflow execution, emphasizing scalability, maintainability, and extensibility.
- Integrated multiple communication platforms including WhatsApp, Reddit, YouTube, and X (Twitter), enabling autonomous content generation, intelligent engagement, and cross-platform task execution.
- Continuing development toward a scalable multi-agent architecture supporting autonomous reasoning, distributed task execution, and intelligent software assistants.

Veriq — AI Framework for Natural Language Driven Software Engineering

2026 – Present

- Designing an AI-powered software engineering framework that transforms natural language software requirements into executable software workflows spanning user interfaces, backend APIs, and system interactions.
- Architected a scalable monorepository comprising independent modules for backend services, React frontend, AI agents, analytics, deployment infrastructure, browser extensions, SDKs, command-line utilities, and third-party integrations.
- Designed backend services using Python and Clean Architecture principles with explicit separation between application, domain, and infrastructure layers to support long-term maintainability.
- Implemented REST APIs, Docker-based development environments, continuous integration workflows, and engineering tooling supporting reproducible development and deployment.
- Designed extensible interfaces enabling future integration of autonomous AI agents, self-healing execution strategies, intelligent planning algorithms, Jenkins plugins, GitHub applications, and multi-language SDKs.
- Produced comprehensive technical documentation including software architecture specifications, API documentation, deployment guides, database design, developer onboarding material, and long-term project roadmap.
- Exploring the application of Large Language Models to software engineering, automated framework generation, intelligent workflow orchestration, and adaptive software maintenance.

E-Commerce Workflow Automation using Playwright

2024

- Developed a modular browser automation framework using JavaScript and Playwright following the Page Object Model design pattern.

- Implemented reusable page abstractions, browser lifecycle management, reporting utilities, and framework organization emphasizing maintainability and scalability.
- Applied object-oriented software design principles to develop reusable components supporting end-to-end customer purchase workflows.

Technical Skills

Programming Languages

Python | Java | JavaScript | TypeScript | C

Software Engineering

Object-Oriented Design | Data Structures & Algorithms | REST APIs | Flask | Django | Robot Framework | Playwright | Selenium

Artificial Intelligence & Data

Large Language Models (LLMs) | Claude API | OpenAI API | Gemini API | Prompt Engineering | OpenCV | scikit-learn | Pandas

Cloud, DevOps & Infrastructure

Docker | Jenkins | Git | GitHub Actions | CI/CD

Databases

SQLite | MySQL

Developer Tools

VS Code | Postman | Jira | GitHub

Relevant Coursework

- Data Structures and Algorithms
- Design and Analysis of Algorithms
- Object-Oriented Programming
- Database Management Systems
- Operating Systems
- Computer Networks
- Computer Organization and Architecture
- Software Engineering
- Software Testing
- Machine Learning
- Artificial Intelligence
- Data Science

Leadership & Extracurricular Activities

- Served as **Creative Head** of **Anveshana**, the college magazine, leading branding, publication design, and visual communication initiatives.
- Led the Creative and Design Team for **FantomCode**, the college hackathon, overseeing event branding, promotional material, and participant-facing design assets.
- Served as **Creative Head** for **RobotFiesta**, directing the visual identity, promotional campaigns, and event design.
- Progressed from Organizing Committee member in the Technical and Design teams to leading creative initiatives for multiple institutional technical events.